

Examining the IC Prediction Market

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The ICPM is a powerful tool that can offer valuable services in a number of key IC functions, including coordination, real-time forecasting, hypothesis-testing, and functions still unthought of.

1. This article is Unclassified in its entirety.

1. Traditionally, analytic judgments have been determined by a relatively small number of analysts in ways unchanged in decades. However, the rise of the interconnected World Wide Web has provided means to permit the Intelligence Community (IC) to make better use of the full capacities of its 20,000 analysts in the assessment process.¹ In this article, one attempt to do so, the IC Prediction Market (ICPM), is studied by students and faculty at the National Intelligence University (NIU), who examined the ICPM since its inception in 2010. Here they present their findings and ponder the ICPM's future, which they argue is full of possibilities.

Leveraging Analytic Potential through Prediction Markets

2. Numerous IC methodologists and analysts have advocated finding ways to enhance the ability of the IC's many analysts to reach one another to test and improve their conclusions. One way to aggregate opinions from a large number of individuals is the use of a prediction market. A prediction market is similar in concept to a traditional stock market, in which traders buy and sell shares of a company. The price of each share goes up or down over

time, and traders profit or take a loss on such movements.²

3. In a prediction market, traders buy and sell shares based on whether they believe an event is more or less likely to occur than the current market price reflects. For example, "Leader X will still be in power of country Y on 31 December 2014." The going market price (based on a 100-point spread) of a prediction reflects the aggregated opinion of how likely an event is to occur at a given point in time. For example, a market price of 34 communicates a collective opinion that there is a 34-percent chance that leader X will still be in power of country Y on 31 December 2014. Over time, the market price rises or falls, and traders profit or take losses on the movements. Eventually, the market on the prediction expires and is closed, either when 31 December 2014 passes or when leader X is ousted.

4. Prediction markets such as the Iowa Electronic Markets, the Hollywood Stock Exchange, Foresight Exchange, and InTrade^a have all operated for many years and

a. InTrade ended its operations in early 2013 in the face of a US Federal Commodities Trading Commission investigation into its trading practices.

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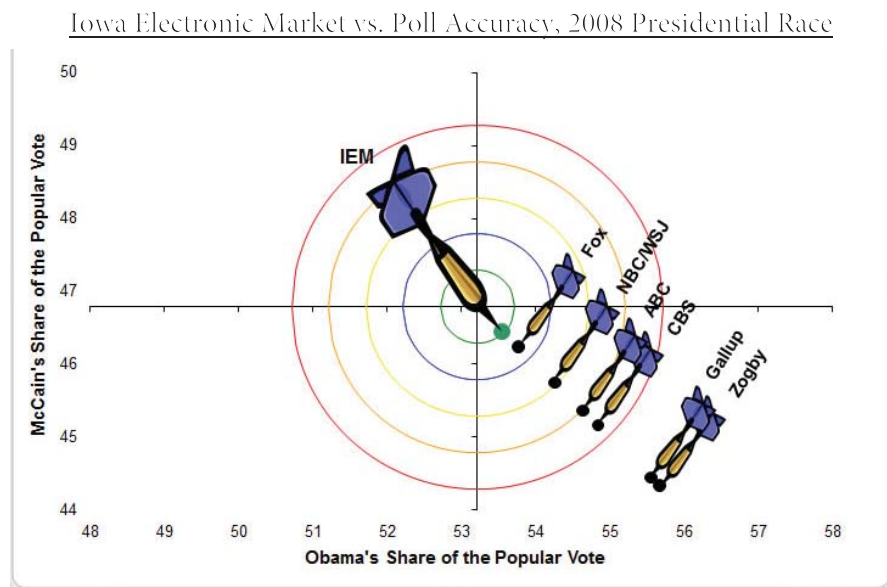
established an impressive record of accuracy.³ Iowa Electronic Markets (IEM), which is run by the faculty of the University of Iowa, for example, opened in 1988. Since then, it has run markets on more than 50 US political campaigns and a handful of foreign political campaigns. It has generally had greater success in predicting outcomes of presidential elections than standard polls.⁴ (See IEM's own assessment of its performance in the 2008 US presidential election in the graphic on right from its website.)

5. In 2010, the Office of the Director of National Intelligence, Analytic Integrity and Standards (ODNI/AIS) established the IC's first prediction market on its classified system to forecast geopolitical events. Over the past 3 years, the number of "traders" in the IC market has grown to more than 2,600, who have traded on some 425 analytic questions. The ICPM does not trade with real money like a traditional stock market. Rather, traders start with a one-time grant of points with which to trade.

6. Daily operation of the ICPM has been handed over to the Intelligence Advanced Research Projects Activity (IARPA) as part of the Aggregative Contingent Estimation Program (ACE).⁵ The market remains jointly managed by IARPA and ODNI/AIS, however.

Evaluating ICPM Data

7. In examining the ICPM, we were interested in comparing the accuracy of the ICPM to the accuracy of public prediction markets. We intuited that the ICPM, whose traders have unique access to classified information, might outdo public prediction markets, where traders did



Graphic courtesy of University of Iowa, Henry B. Tippie College of Business—<http://tippie.uiowa.edu/iem/media/08pres.html>.

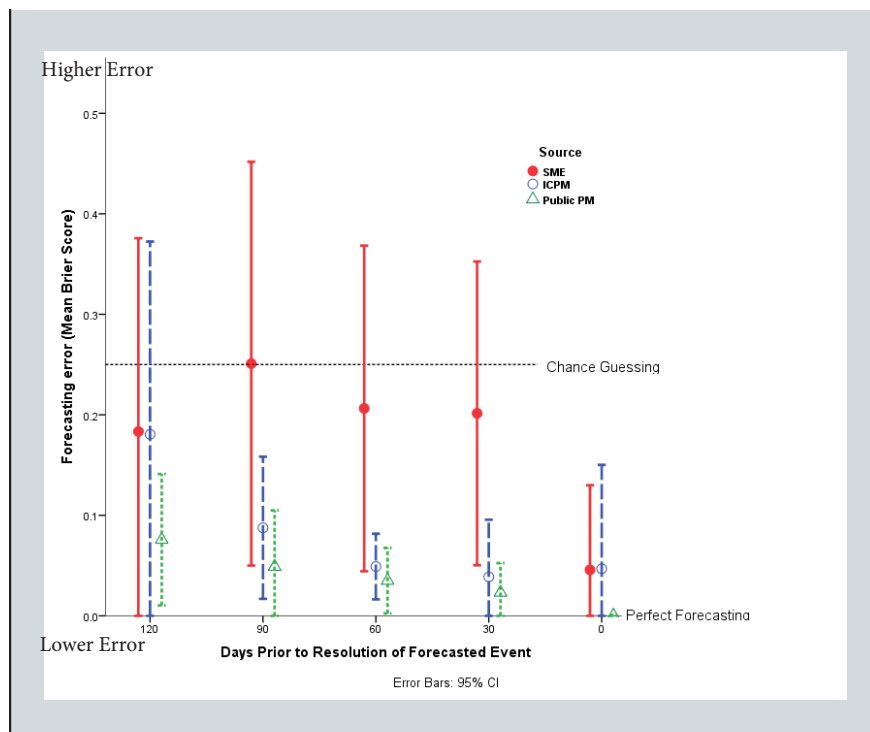
not have access to classified information. The public prediction market we examined was InTrade, which had marketed predictions that were similar to those IC analysts were being asked. For example: "Muammar al Gaddafi to no longer be leader of Libya before midnight 31 December 2011," "Iran to conduct a nuclear weapons test on/before 31 December 2011," and "Hugo Chavez to no longer be the President of Venezuela before midnight 31 December 2011." We identified 10 such predictions and purchased the trading data from InTrade. We replicated some of these markets in the ICPM, or identified closely analogous predictions if they existed, so that direct comparisons between the two prediction markets could be made over time.

8. We were also interested in comparing the accuracy of the prediction market to what is often the norm in the IC, assessments made by single (or a few) subject-matter experts.

Based on studies done in other domains, we expected the market forecasts to be better than those of single experts. We took 10 questions from the ICPM and found the 10 best IC experts we could identify on each topic. Our experts had, on average, 19.6 years of intelligence experience and 13.3 years of experience working their current portfolios.

9. We asked our experts to evaluate the likelihood of an event on a scale of 0 to 1, where "0" corresponded to "Impossible" and "1" corresponded to "Certainty."⁶ Forcing our experts to make explicit forecasts like this enabled us to directly compare their forecasts with market outcomes over the same time period.

10. We repeatedly collected forecasts from our markets and our experts to sample various time horizons, ranging from very near-term forecasts to as long as 4 months before a subject was resolved. All told,



we collected 152 individual forecasts from the ICPM, InTrade, and individual IC experts over approximately matching topics and time horizons.⁷

11. Once all the markets had been resolved and we knew the outcomes, we compared the forecasting accuracy of the ICPM, InTrade, and IC experts.⁸ We found, as expected, the market prices provided by the two prediction markets provided significantly more accurate forecasts than the ones provided by our experts.⁹ Counter to our hypothesis, we did not see a statistical difference in accuracy between the ICPM and InTrade.¹⁰ (See table above.)

Why Did the ICPM Behave as it Did?

12. In our sample, as in other studies, forecasts made by prediction markets were more accurate than forecasts made by single individuals. Still, we wondered if some of the

negative aspects of a market system might also be present in the ICPM. Could the ICPM be subject to price manipulation schemes, bubbles, flash crashes and the sort? What were the motivations of the traders in the ICPM? What exactly is driving change in market prices? Other studies suggest a variety of theoretical explanations for market price change: Hayek's Hypothesis, Efficient Market Hypothesis, Random Walk Theory, and Condorcet's Jury Theorem all have something to say on the topic.¹¹

13. Rather than attempting to refine the considerable amount of theoretical work already done on this topic, the authors decided on a more practical approach of interviewing the IC traders themselves and asking them why they chose to buy and sell predictions in the ICPM. Our results did not contradict previous theoretical work, but they did give some helpful insight into the mechanics of the ICPM.

14. We interviewed the dozen traders who had accumulated the highest point totals in the ICPM. These traders represented some of the most active, most successful traders in the ICPM. Questions we asked centered on the amount of research each trader did, how much information sharing they engaged in, what their motivations were, and what strategies they employed. On the basis of these interview results, we developed a survey and administered it to a larger sample of 60 ICPM traders. Forty-four responded.¹²

Research

15. Most traders used a mixture of classified and unclassified research, most heavily weighted toward open source (OSINT) material. The survey showed that new OSINT was the most common cause for a sudden price change. Only four interviewees said they used classified information not readily accessible to all IC analysts through the main web pages of NSA, CIA, NSA, or DIA.

Exchange of information

16. Comments in the ICPM web pages were the second most-mentioned reason given for price changes. IC market participants regularly used the comments feature of the software to take note of important new intelligence or open-source information. This clearly had an effect on prices.

Mechanics of the market

17. A third reason given for price changes had to do with peculiarities in the mechanics of the prediction market, especially when a market first opened. When a predication is first offered, the opening price is set at a neutral value of 50 (50-percent probability), and traders take ad-

vantage of the market inefficiencies present at the initial public offering.

18. Similarly, mechanics of the market affected the prices as a prediction market reached an expiration date. As that date neared, prices drifted toward an amount that reflected a continuation of the status quo. This is perfectly reasonable since, as time grew short and a prediction hadn't occurred, there was less likelihood that it would. Traders would also liquidate some shares when the price tended to stagnate near 100 percent or 0 percent, because less opportunity for price movement meant less opportunity for profit. Prices nearer the 50-percent level promise more volatility and more opportunity for profit.

Taxonomy of trader behavior

19. The feedback from ICPM traders suggested that trader motivations and behavior could be classified.

20. The expert trader. Some traders cited forecasting accuracy as an objective. These traders focused on a limited number of issues. Narrowing the focus allowed them to do more research per prediction, so traders tended to be more knowledgeable about each topic.

21. These “expertise-centered traders” often dealt with questions that were directly related to the accounts they worked on daily. Point accumulation was not an expert trader's major goal, and so they were not concerned about making the most of their limited trading resources. They tended to buy and hold shares for longer periods, even for predictions with little volatility (and little potential for profit).

22. Point-centered traders. These traders viewed the ICPM as a game to be won by accumulating points from the markets. These traders entered a larger number of markets and traded more often than the expert traders. They looked to take advantage of market inefficiencies to maximize profits, and when profit-potential waned, they liquidated their holdings and moved on. Occasionally they bet on a long shot since they were most interested in what would generate profit, rather than what was most likely to happen.

23. The survey indicated that the above dichotomy in trading behavior really represents a spectrum of behavior, rather than a pure binomial classification. Most traders exhibited a mixture of both expertise-centered and point-centered motivations and behavior, with a relatively small number gravitating entirely toward one of the two extremes.

24. It's not certain that one type of trader is more valuable to market accuracy than the other. A market of expert traders might, at first blush, seem a positive thing. However, expert traders build up fewer points and tend to have less influence on market outcomes.

25. At the same time, the point-centered traders are the ones most responsible for quickly elimi-

nating market inaccuracies after an initial public offering or after the appearance of news that may suddenly and sharply affect prices. In short, it seems that both types of traders are required for the best market predictions.

The nature of the predictions

26. We also wanted better understanding of the behavior of each of the predictions marketed in the ICPM. Would market behavior suggest, for example, that some questions were more easily answered than others?

27. One intuition we had was that prediction markets concerning high-priority issues—as defined by the National Intelligence Priorities Framework (NIPF)¹³—might lead to more accurate forecasts than predictions concerning low-priority issues. According to our reasoning, predictions concerning higher priority issues were more apt to have a greater proportion of collection assets focused on them and a larger number of analysts following them.

28. Such increased resources ought to translate to increased accuracy, we thought.¹⁴ Not so, according to our sample. We examined 271 resolved predictions from the ICPM and coded each one according to the NIPF issue it addressed. We found no

Figure 2. Trader motivations and behaviors

Point trader	Expert trader
Motivated by points	Motivated by market accuracy
Lower-level of expertise per contract	Higher-level of expertise per contract
Larger number of contracts traded	Smaller number of contracts traded
Higher number of trades	Lower number of trades
Less research time	More research time
Higher points total	Average points total
Concerned with behavior of other traders	Not concerned with other trader behavior

correlation between the NIPF priority and the accuracy of prediction.¹⁵

29. Although the NIPF is supposed to focus collection and analytic resources, we were unable to support, through the outcomes of ICPM trading, the notion that such concentration of assets leads to improved predictions in the ICPM. Still, the existence of the ICPM at least makes it possible to examine such a question empirically. Additional research along these lines may help the IC further understand what kinds of predictions the IC is best at providing.

30. Given that the ICPM was able to predict low-priority issues about as accurately as high-priority ones, one application of the ICPM might be to use it to address lower-priority intelligence issues not already addressed by IC flagship products. On the other hand, there is a limit to how many questions the ICPM can handle. A finite number of traders cannot trade on an infinite number of questions, so some restrictions based on priority might eventually be required.

Pondering the ICPM's Future

31. The ICPM has operated on a primarily experimental basis now for more than three years. Perhaps the time has come for the IC to decide what it wants of its prediction market and consider more operational applications for it. Identifying the desired purpose is important, as different purposes will require different approaches to its structure and operation. Following are a few suggestions and their implications.

Purpose 1: Support to decisionmakers

32. Most apparent, the ICPM forecasts could be used to inform

decisionmakers. Unlike static intelligence products, the ICPM market price continually updates, providing a quick summary of prevailing opinion as well as a barometer alerting observers to changes in opinion.

33. A portfolio of current market prices could be used to populate an efficient “cockpit display” for strategic decisionmaking.¹⁶ One can imagine a customized screen on a mobile device that, at a glance, shows a policymaker the current analytical opinion on a variety of questions of interest to the owner of that tablet. Such an interface might show decisionmakers supporting evidence/events overlaid on the price chart, along with levels of confidence derived from trading activity, along with indications of the historical accuracy of the market in addressing similar issues.

34. Decisionmakers should be permitted to inject new topics for prediction into the market. This will require fuller understanding, both in the IC and among policymakers, of what predictions can and cannot be made to work in a market. Predictions must be carefully scoped and worded so there is little ambiguity surrounding their meaning and what constitutes an endpoint for that prediction. Otherwise, controversies ensue over “what really happened.” This requires a clear time frame, clear and consistent definitions, and predictions that can eventually be verified with objective, ground truth.

35. A weakness here is that prediction market questions are often simplified in ways that eliminate the context decisionmakers often desire. For example, take this question: “How likely is a major conflict between North Korea and

South Korea?” To put this question to the ICPM requires a clear definition of what constitutes a major conflict and a reasonable time frame within which it would or would not occur. The ICPM version of the question might be: “Will a major conflict take place between North Korea and South Korea before 1 January 2015?” The question would include a specific definition of a “major conflict.” While this market might provide value, its use of a somewhat arbitrary definition of “major conflict” limits its scope by not addressing the potential for conflict beyond 1 January 2015 and by not addressing the fuller, richer picture of the nuances, drivers, and consequences of a potential conflict.

36. The ICPM attempts to overcome such shortfalls. Conditional questions (e.g. “If X occurs, then how likely is Y?”) can help provide probabilities of specific consequences or even the consequences of a specific US action. Multiple choice questions can help narrow date ranges and the potential scale of the conflict. Nonetheless, the reductionism required may make participants feel as though they are examining the pixels of a greater intelligence picture. Although such constraints mean the ICPM will never be able to answer all decisionmaker needs, we still believe the ICPM can provide a valuable service for a large set of intelligence questions.

Purpose 2: Support to analysts

37. The ICPM can serve to marshal analytic help. Analysts outside the Beltway sometimes have difficulty getting IC help in addressing their often second-tier issues. By injecting questions into the IC market, a distant analyst can hope to obtain ana-

lytic insight beyond what is readily accessible far from Washington.

38. The ICPM provides a mechanism to tap analytic expertise that might not be specifically assigned to the subject of the question. An analyst with a question about country X would naturally reach out to the people who work the country X account. But what of those who have expertise on country X but are not assigned to the desk? Furthermore, expertise could come from areas that face issues like the one in country X. In effect, the ICPM can engage analytic knowledge no matter where it resides on an organization chart.

39. Because the ICPM comment capability facilitates crosstalk, including the sharing of evidence and reasoning, the market can serve as a collaborative tool that bypasses normal organizational hindrances.

40. The recent addition of a prediction market on SIPRNET will give analysts outside the beltway greater access to intelligence analysis. Intelligence personnel deployed in forward locations are immersed in their accounts in ways those inside the Beltway are not, and their tactical knowledge of events in their regions of responsibility is, in many ways, superior. Integrating such analysts into the ICPM gives them greater voice to affect IC assessments and to enlarge the knowledge of those inside the Beltway.

41. The ICPM offers an efficient means of capturing degrees of uncertainty on given topics, historically a difficult task. "Is this question really uncertain," a decisionmaker may ask, "or do you just not know what you are talking about?" A market price of 50 percent reflects more uncertainty

than a price of 10 percent or 90 percent. With such relatively concrete expressions, collectors and analysts can better focus their efforts.

Purpose 3: Identification of the best forecasters in the IC

42. A prediction market is one way to aggregate opinion, but it is not the only way. IARPA's ACE program has experimented with a variety of methods to elicit, aggregate, and communicate forecasts.¹⁷ Very promising results have been obtained by identifying the best forecasters from a group and then mathematically combining individual forecasts weighted according to various criteria such as experience or past performance. Imagine a "super unit," composed of the best the IC has to offer, brought together to make forecasts on key issues. Initial results suggest such an approach provides more accurate forecasts than other methods, including forecasts elicited from a prediction market.

43. Because the ICPM identifies "winners"—the traders who profit the most—it is tempting to think that the market could be used to identify the IC's best forecasters. Our bottom line is that using it for that purpose would be inappropriate. From our interviews, we know the most successful traders are point-traders who take advantage of market inefficiencies, not expert knowledge, to accumulate points.

44. But is the effective gamer the kind of forecaster the IC would want to identify? Moreover, forecasting accuracy is only one component of what the IC and US policymakers require of their intelligence analysts.¹⁸ Nonetheless, the ICPM does provide a framework to consistently observe analytic accuracy in ways that may

be useful to IC leaders, policymakers, and analysts.

Purpose 4: Testbed for study

45. For intelligence to improve as a discipline, we must advance the theory of intelligence. We cannot view each intelligence question as a unique and beautiful snowflake. To advance the field, we must identify relationships and tools that can be generalized and are useful across a broad range of time periods, circumstances, and locations. As we increase our understanding of such relationships, the theory of intelligence advances and the practice of intelligence improves. In addition, this process requires a testing ground in which hypotheses and methods can be tested and evaluated. We believe the ICPM provides a neutral, objective, and data-rich arena in which theories and tools can be tested and compared.¹⁹

46. The ICPM could also be used to examine the structured analytic methods that have become common throughout the IC. For example, the trading program could be altered to allow association of trades with a particular methodology and thus provide data for an empirical examination of methods. A researcher might decide to make trades based on the result of a Bayesian belief network. Another set of questions might be traded based on a matrix of competing hypotheses, an alternative futures analysis, or a poll of experts. Over time, data could reveal the most effective methods, and insights gained could lead to refinement of methods.

In Sum

47. The ICPM need not be confined to a single purpose; it can serve all of the purposes described above.

And, although the requirements of intelligence analysts and policymakers exceed the services the ICPM can provide, the market is nevertheless a powerful tool that can offer valuable services in a number of key IC functions, including coordination, real-time forecasting, hypothesis testing, and functions still unthought of.



Endnotes

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